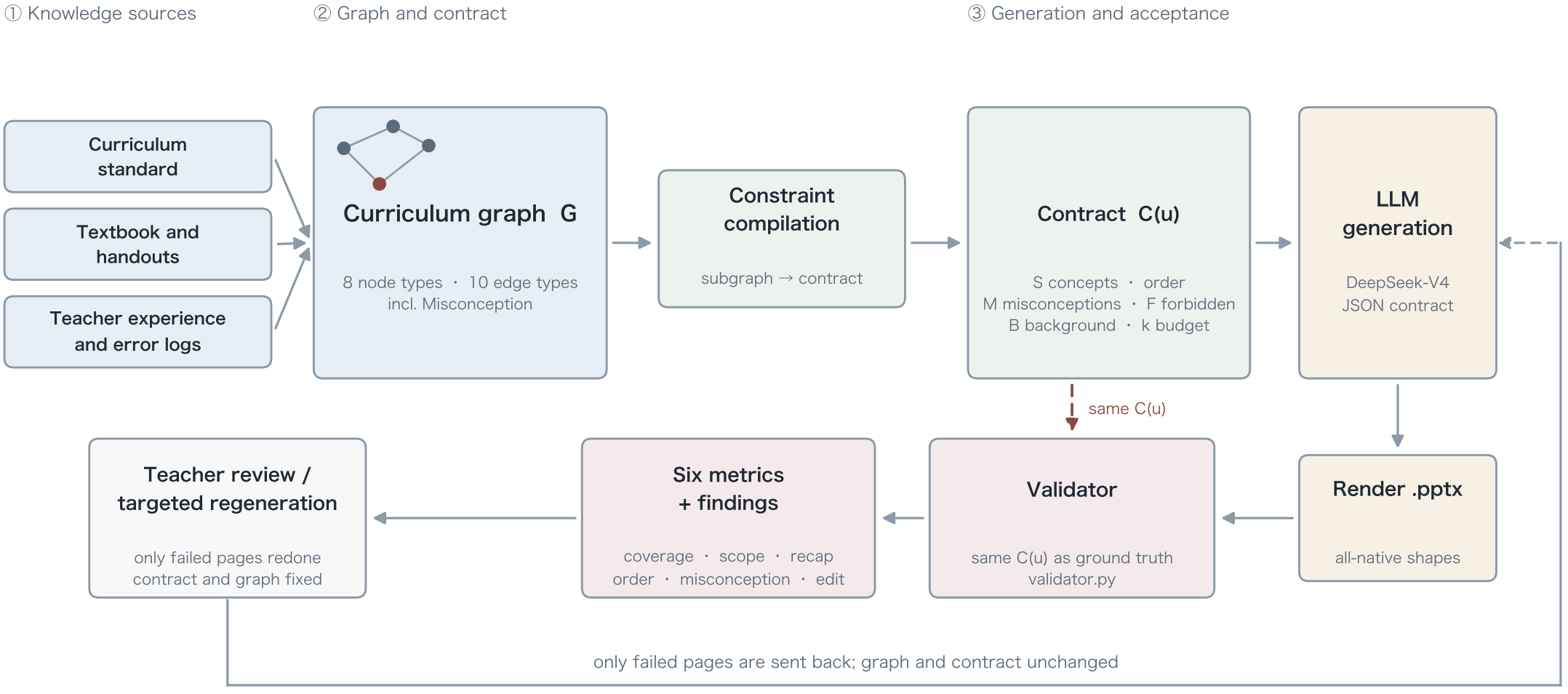
① Knowledge sources
② Graph and contract
③ Generation and acceptance
Curriculum standard
Textbook and handouts
Teacher experience and error logs
Curriculum graph G
8 node types · 10 edge types
incl. Misconception
Constraint compilation
subgraph → contract
Contract C(u)
S concepts · order
M misconceptions · F forbidden
B background · k budget
LLM generation
DeepSeek-V4
JSON contract
same C(u)
Render .pptx
all-native shapes
Validator
same C(u) as ground truth
validator.py
Six metrics + findings
coverage · scope · recap
order · misconception · edit
Teacher review / targeted regeneration
only failed pages redone
contract and graph fixed
only failed pages are sent back; graph and contract unchanged